

Urban Air Quality and Respiratory Health: A Five-Year Cohort Study

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Abstract

Background: Chronic PM_{2.5} exposure is linked to elevated asthma and COPD rates. This study examines PM_{2.5} concentrations and emergency-department (ED) visit rates across 48 urban census tracts over five years (2018–2022). A 10 µg/m³ increase in annual mean PM_{2.5} was associated with a 12.4% increase (95% CI: 8.7–16.2%, p<0.001) in respiratory ED visits. Low-income tracts showed 1.8× greater effect size.

1. Introduction

Ambient air pollution causes ~4.2 million premature deaths annually (WHO, 2021). PM_{2.5} is the most harmful component. Urban environments concentrate emission sources including vehicular traffic, industrial activity, and domestic heating.

2. Methods

Study population: 48 census tracts, population 2,100–8,900. Air quality data from 12 EPA-certified monitoring stations interpolated by inverse distance weighting. Panel fixed-effects regression with heteroskedasticity-robust standard errors clustered at tract level. All analyses in R 4.2.1 (plm package).

3. Results

Mean annual PM_{2.5} ranged 6.2–28.7 µg/m³ (grand mean 14.3). ED visit rates: 87–412 per 10,000 person-years. Stratified analysis: 9.1% effect in high-income tracts vs 17.6% in low-income tracts.

4. Conclusions

Targeted air quality interventions in high-exposure, low-income tracts may yield disproportionate health benefits. Estimated cost saving: \$2.7M/year against an intervention cost of \$800K–\$1.2M for a 5 µg/m³ PM_{2.5} reduction.